

Project for the Reduction and Control of Non-Revenue Water (NRW) in Escalante Water District, Escalante City, Negros Occidental Province, Philippines

Republic of the Philippines

Project Concept Paper

February 2026

<i>Applicant Information</i>	
Name	
Position	
Organization	
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PROJECT/ PROGRAM CONCEPT PAPER (PCP)

SECTION 1. BASIC PROJECT INFORMATION		
1.1	Country	Republic of the Philippines
1.2	Title	Project for the Reduction and Control of Non-Revenue Water (NRW) in Escalante City, Negros Occidental Province, Philippines
1.3	Location(s)	Escalante City, Negros Occidental, Philippines
1.4	Duration	180 months (2027-2041)
1.5	Budget (total)	1,726,400.00 USD (Grant from Ministry of Environment)
1.6	Objectives	To reduce non-revenue water in the Escalante Water District by addressing physical losses through targeted system improvements, thereby improving the efficiency, reliability, and sustainability of water supply services.
1.7	Beneficiary	Sustainable and efficient water service delivery for the 97,050 residents of Escalante City.
1.8	Implementing organization	Ministry of Environment, Republic of Korea
	- Type	<i>Please check the type of the implementing organization in the boxes.</i> ☐ Central/national government ☐ Provincial government ☒ Public organization ☐ Others.....
	- Major Functions	The primary function of this project is to reduce Non-Revenue Water (NRW) for the citizens of Escalante City, which is primarily caused by leakages. We aim to optimize the water supply system of Escalante Water District by replacing the mainline pipes, service connections, and customer meters to reduce leakage volume.

SECTION 2. PROJECT RATIONALE

SITUATION ANALYSIS: Please provide a brief introduction to the current social and economic situation related to the Project (geographic region and beneficiaries, etc.)

Please describe the problem or critical issue which the project seeks to resolve, how the problem was identified, and how will the Project address the problem. If relevant, analysis on gender equality needs to be described.

Escalante City is a fourth-class city located in the Province of Negros Occidental in the Western Visayas Region of the Philippines. The city is bounded on the north and west by the City of Sagay, on the south by the Municipality of Toboso, and on the east by the waters of the Tañon Strait. Water supply services in the city are provided by the Escalante Water District (EWD), which is responsible for the operation and maintenance of the existing water supply system and the delivery of potable water to residential, commercial, and institutional users.

Based on the records of EWD and following a detailed investigation of the water supply system using the International Water Association (IWA) Water Balance methodology, the factors contributing to total Non-Revenue Water (NRW) have been quantified. Results indicate that 99.61% of NRW is attributable to physical losses due to leakages, while 0.08% is due to unbilled authorized consumption and 0.31% is caused by unauthorized consumption, meter inaccuracies, and data handling errors. These findings clearly indicate that NRW in the Escalante Water District is overwhelmingly driven by physical system deficiencies rather than commercial or administrative losses.

The water loss performance of the Escalante Water District is classified as Category D under the IWA performance categories. This category represents a very defective technical approach to water system management, characterized by high leakage rates, aging and deteriorated pipeline infrastructure, insufficient pressure management, and limited capacity for systematic leak detection and repair. Systems falling under this classification typically experience elevated operating costs, inefficient use of water resources, and increasing risks to service reliability over time.

Without targeted interventions, continued physical losses are expected to worsen as system components further deteriorate, leading to higher production and pumping costs, increased stress on water sources, and reduced ability of the utility to meet growing demand. Persistent NRW also constrains the financial capacity of the water district to reinvest in infrastructure improvements, creating a cycle of deferred maintenance and declining system performance.

The proposed Non-Revenue Water Reduction Project of the Escalante Water District is technically and economically feasible based on the conducted economic analysis. Implementation of the project is projected to result in cumulative water savings of approximately 7.49 million cubic meters over a 15-year period, water that would otherwise be lost through system leakages. The estimated Economic Internal Rate of Return (EIRR) of 24.3% exceeds the NEDA-prescribed social discount rate of 10%, indicating strong economic justification. Furthermore, the project yields a Cost-Benefit Ratio (CBR) of 3.04, reflecting substantial returns relative to investment costs, and generates an estimated net economic benefit of PHP 273.74 million for the City of Escalante.

Through systematic replacement of deteriorated pipelines, upgrading of service connections, and installation of accurate customer meters, the project will address the root causes of physical water losses while strengthening the long-term operational sustainability of the water supply system. These interventions are expected to improve water use efficiency, enhance service reliability, and support the continued delivery of safe and adequate water to the residents of Escalante City.

2.2	COUNTRY DEVELOPMENT STRATEGIES AND POLICIES: Please describe how the Project related to other relevant national development strategies and policies, and provide the ongoing status of their implementations, results and effects, if any.
	1. National Medium-Term Development Plan (The Core Framework) This is the highest-level national development plan and serves as the basis for all (ODA) projects. Title: Philippine Development Plan (PDP) 2023-2028 Published by: National Economic and Development Authority (NEDA) Goal: Achieve “Upper Middle-Income Country” status and reduce the poverty rate to single digits (9%) by 2028 Strategy: Targets a complete economic and social transformation. It explicitly lists “Expand and Upgrade Infrastructure” as one of its cross-cutting strategies. 2. Key Infrastructure Policy (Infrastructure Flagship Program) This is the current administration’s flagship infrastructure program, expanding upon the previous “Build, Build, Build” initiative. Title: Build Better More (BBM) Program Lead Agencies: Department of Public Works and Highways (DPWH), Department of Transportation (DOTr), etc. Goal: Going beyond simple construction, it emphasizes “Better” quality and digital/physical connectivity. It comprises over 190 infrastructure Flagship Projects (IFPs), including 44 Water Resources projects (totaling approximately 839.1 billion PHP). 3. Long-term National Vision This vision outlines the long-term national goals leading up to 2040. Title: AmBisyon Natin 2040 Published by: National Economic and Development Authority (NEDA) Goal: all Filipinos to enjoy a “strongly rooted, comfortable and secure life” (<i>Matatag, Maginhawa, at Panatag na Buhay</i>). Access to safe and clean water is considered a fundamental prerequisite for a “Comfortable Life (<i>Maginhawa</i>).” 4. Latest Legislative/Policy Trends (Water Sector Specific) This is the most urgent water-related legislation currently being pushed by the Philippine Congress and Administration. Title: Department of Water and Resources (DWR) Act (Proposed creation of the DWR) Status: Discussed as a priority bill in both the Senate and the House (ongoing as of 2025). Goal: Aims to establish a “Department of Water Resources” to integrate water management functions currently fragmented across various agencies (NWRB, LWUA, DPWH, etc.).

JUSTIFICATION FOR INTERVENTION: Please describe how the need for the Project was determined, and what the rationale/justification for the Project (why the Project is considered to be the most effective way the problem is resolved).

1. The need for the project was determined based on the results of a comprehensive feasibility study and assessment of the existing water supply system of Escalante Water District, which identified a non-revenue water level of 62.29%, with approximately 99.61% of these losses attributed to physical leakages. The study further indicated that aging and deteriorated mainlines, service connections, and customer meters also contribute to the water losses, resulting in reduced system efficiency, unreliable water supply, and financial losses for the utility.
2. Addressing these problems improves water availability for consumers, and strengthens the operational and financial sustainability of the water district. NRW reduction offers immediate recovery of lost water, maximizes the use of existing infrastructure, and supports long-term service reliability.

Water Balance Status in the City of Escalante, 2020

System Input Volume 1,258,116.87	Authorized Consumption 475,037.48	Billed Authorized Consumption 474,463.00	Billed Metered Consumption 474,463.00	Revenue Water 474,463.00
			Billed Unmetered Consumption 0.00	
		Unbilled Authorized Consumption 574.48	Unbilled Metered Consumption 0.00	Non-Revenue Water (NRW) 783,653.87 62.29% of SIV
			Unbilled Unmetered Consumption 574.48	
	Water Losses 783,079.39	Apparent Losses 2,444.27	Unauthorized Consumption 1,258.12	
			Customer Metering Inaccuracies and Data handling errors 1,186.16	
		Real Losses 780,635.11 99.61% of NRW		

3. The replacement of deteriorated mainlines, service connections, and customer meters represents the most effective strategy for reducing non-revenue water in the system, as it directly addresses the primary sources of physical losses. Mainline replacement reduces high-volume leaks in the distribution network, service connection replacement eliminates widespread and often undetected household-level leakages, and customer meter replacement ensures accurate measurement and monitoring of water consumption. Given that approximately 99.61% of the existing NRW is attributed to physical losses, this integrated approach provides the most direct and sustainable solution to the high NRW problem.

SECTION 3. PROJECT DESCRIPTION

Objective/Outcome/Output: Please outline the objectives, the expected outcomes, and output of the Project.

(Objectives) To reduce non-revenue water in the Escalante Water District by addressing physical losses through targeted system improvements, thereby improving the efficiency, reliability, and sustainability of water supply services.

(Output) Replacement and rehabilitation of deteriorated water transmission and distribution systems mainlines in identified high-loss areas of the Escalante Water District.



Activities: Please describe what will be carried out in terms of planned activities, their timing and duration, and who will be responsible for each activity. It should indicate the sequence of all major activities and implementation milestones.

1. NRW Reduction Strategies: Replacement of Mainlines, Service Connections, and Customer Meters (2028-2031)

- This phase focuses on achieving a substantial and measurable reduction in NRW through targeted infrastructure replacement. Given that 99.61% of NRW is caused by physical losses, the project prioritizes the replacement of deteriorated mainlines, service connections, and customer meters as the most effective and sustainable intervention. Pipe and connection replacement addresses recurring leaks at their source, while meter replacement ensures accurate measurement of water consumption. Implemented over four years, these interventions provide a long-term solution to NRW reduction by minimizing future leak recurrence and improving overall system efficiency.

- This component ensures the long-term sustainability of NRW reduction gains by strengthening system reliability and maintenance practices. Key activities include the replacement of supply and production meters, establishment of District Metered Areas (DMAs), active leak detection and repair, and pressure management. These measures enable continuous monitoring of system performance, early identification of new leaks, and controlled operating pressures that reduce pipe stress. Together, they prevent the recurrence of high NRW levels and support uninterrupted, efficient system operation beyond the reduction phase.

- This strategy strengthens institutional capacity and supports data-driven water utility management. Capacity building programs will enhance the technical skills of water district personnel in NRW management, system operation, and maintenance. GIS mapping and hydraulic modeling will provide accurate system data, support planning and decision-making, and improve monitoring of network performance. These non-engineering measures ensure that physical investments are effectively managed and sustained over the long term.

[illegible]

3.3	Budget: Please provide a brief requirement by Outputs and Activities. The budget requirement is not full and well-defined. The detailed budget requirement will be fully elaborated after the PCP is selected. This information is to better understand the project.		
	Output	Activity	Proposed budget (in USD)
	NRW Reduction	Mainline Replacement	1,000,000.00 USD
		Service Pipe Replacement	221,000.00 USD
		Customer Meter Replacement	141,000.00 USD
		Establishment of DMA	31,000.00 USD
	Operation and Maintenance	Supply/Production of meter replacement	20,000.00 USD
		Leak Detection Equipment	50,000.00 USD
		Stored Materials	32,000.00 USD
	Non-Engineering Interventions	Freight and Handling	1800.00 USD
		Capacity Building	10,700.00 USD
		GIS Mapping	1400.00 USD
		Hydraulic Modeling	100.00 USD
		Physical Contingencies	216,000.00 USD
	Total		1,726,400.00 USD

SECTION 4. STAKEHOLDER ANALYSIS

	TARGET BENEFICIARY: Please describe the following information: a) direct and indirect/wider beneficiary group, b) number of beneficiaries, with gender segregation if necessary (e.g. 300 children rather than children in 3 schools), c) how the target group was identified, d) why they were selected as target group, e) how intended beneficiaries have been involved in Project design, and their expected role in Project implementation and evaluation. If relevant, the target group needs to be disaggregated by sex.
4.1	- Direct Beneficiaries: The main beneficiaries are the residents of Escalante City, who will have reliable, 24/7 access to safe water throughout the city's distribution system. This group includes households, educational institutions, healthcare facilities, and small local businesses that currently experience disruptions in water supply or face water quality challenges. By improving access and reliability, the project will directly enhance daily living conditions, public health, and operational efficiency for these stakeholders. - Indirect/Wider Beneficiaries: The project will also benefit local industries, agricultural enterprises, and future urban developments that depend on a stable and sufficient water supply. Additionally, public institutions and future residents will benefit from the project's sustainable water management practices, which support long-term economic growth, environmental stewardship, and resilient urban planning in Escalante City.
4.2	OTHER STAKEHOLDERS: Please describe other stakeholders (e.g. partner government agency, international organization, NGO, donor agency, etc.), if any, including a) name.group, b) respective role(s) and cooperation/coordination mechanism, etc. In order to smoothly carry out the project, the coordination of the related authorities such as Local Water Utilities Administration (LWUA), Korea Water Resources (K-water), Local Communities and Residents and Local Industries are essential not only for collection of any data and documents required to prepare and implement the project, but also for support and assistance for administrative procedures, permission and etc.

SECTION 5. PROJECT MANAGEMENT AND IMPLEMENTATION

	PROJECT MANAGEMENT Please describe a) who will be responsible for planning and management of the Project operations as well as coordinating other bodies and organizations associated with the Project, b) what arrangements will be established to ensure that there will be effective coordination with other relevant programs and activities.
5.1	The executing organization of the Project is the Ministry of Environment (MOE) of the Republic of Korea. As the entrusted implementing organization, Korea Water Resources Corporation (K-water) will thoroughly manage the overall project. This project will be mainly conducted by experienced Korean contractors. Local Water Utilities Administration (LWUA) will support administration measures, personnel, information and permits for the successful implementation of the project.

